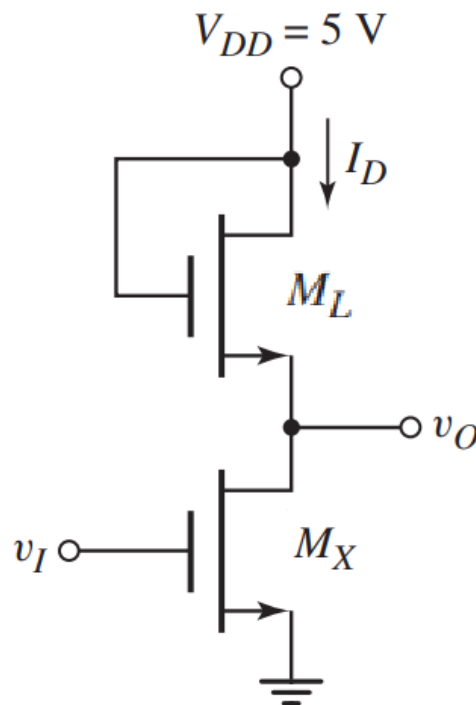


NMOS with Load

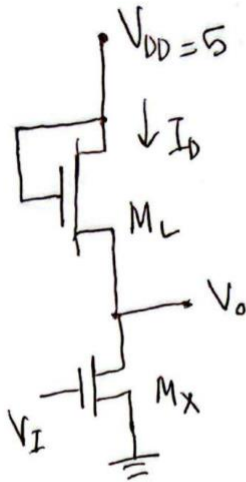
Question 1

For the circuit below (enhancement Load Inverter), assume these parameters, MOSFET process parameter for M_L is $K'_L = 5 \mu\text{A}/\text{V}^2$, $(W/L)_L = 2$ and threshold voltage, $V_{TNL} = 0.7\text{V}$. MOSFET process parameter for M_X is $K'_D = 25 \mu\text{A}/\text{V}^2$, $(W/L)_D = 4$ and threshold voltage, $V_{TND} = 0.7\text{V}$.



(a)	Assume $V_I = 5 \text{ V}$. Find value of V_O in volt. Find I_D and Power Dissipation. Verify the operating mode of M_L and M_X .
(b)	What should be the value of V_I if we want M_L to operate in the triode region?

Solution:



(a) $V_I = 5$

$$V_{TNL} = 0.7$$

$$V_{TND} = 0.7$$

$$k_L = k'_L * \left(\frac{W}{L}\right)_L = 10 \mu A/V^2$$

$$k_D = k'_D * \left(\frac{W}{L}\right)_D = 100 \mu A/V^2$$

For $M_L \rightarrow V_{uDL} = \cancel{1.8} \cancel{7} 5 - 5 = 0 < V_{TNL}$
(0.7)
 $\therefore M_L$ in Saturation

For $M_X \rightarrow V_{uDX} = V_I - V_O = 5 - (\text{small value})$

$\therefore V_{uDX}$ should be greater than V_{TND} (0.7)

$\therefore$ Assume M_X in Triode.

$$\therefore I_D =$$

$$k_L [(V_{GS_L} - V_{TNL})^2] = k_D [2(V_{GS_X} - V_{TND})V_{DS_X} - V_{DS_X}^2]$$

$$\hookrightarrow 10 [5 - V_o - 0.7]^2 = 100 [2(5 - 0.7)V_o - V_o^2]$$

$$\hookrightarrow V_o = \begin{cases} 8.399 \rightarrow V_o \neq V_{DD} \\ 0.2001 \rightarrow \text{Ans} \end{cases}$$

$$\therefore V_o = \boxed{0.2001 \text{ V}}$$

$$\begin{aligned} \therefore I_D &= k_L [V_{GS_L} - V_{TNL}]^2 = (10 \mu\text{A}) [5 - 0.2001 - 0.7]^2 \\ &= \boxed{0.168 \text{ mA}} \end{aligned}$$

$$\begin{aligned} \therefore \text{Power} &= I_D (V_{DD} - 0) = (0.168 \text{ mA}) (5 \text{ V}) \\ &= \boxed{0.840 \text{ mW}} \end{aligned}$$

Verify $M_L \rightarrow$

$$\cancel{V_{GD_L} = 5 - V_o = 5 - 5 = 0}$$

$$V_{GD_L} = 5 - 5 = 0 < V_{TND}$$

$\therefore M_L$ in saturation

Verify $M_X \rightarrow$

$$V_{GD_X} = 5 - V_o = 5 - 0.2007 = 4.8 \text{ V} > V_{TND}$$

$\therefore M_X$ in Triode.

(b) For $M_L \rightarrow$

$$V_{GD_L} = V_{DD} - V_{DD} = 0 < V_{TND}$$

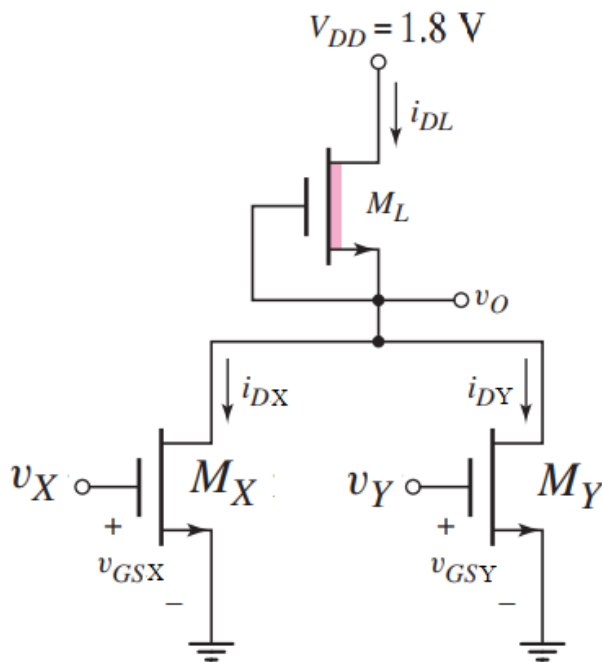
The Gate is connected to Drain.

$\therefore M_L$ cannot be in Triode mode.

Question 2

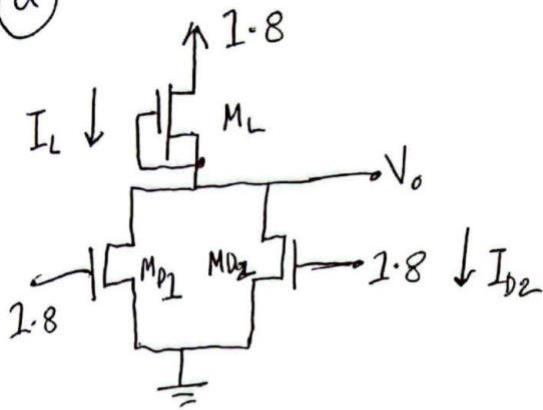
For the circuit below (Depletion Load NOR), assume these parameters, MOSFET process parameter for M_L is $K'_L = 50 \mu\text{A}/\text{V}^2$, $(W/L)_L = 2$ and threshold voltage, $V_{TNL} = -0.6\text{V}$. MOSFET process parameter for both M_X and M_Y is $K'_D = 33.25 \mu\text{A}/\text{V}^2$, $(W/L)_D = 4$ and threshold voltage, $V_{TND} = 0.4\text{V}$.

(a)	Assume $V_X = V_Y = 1.8\text{ V}$. Find value of V_O in volt. Find I_{DX} , I_{DY} , I_{DL} and Power Dissipation. Verify the operating mode of M_L .
(b)	Now, assume $V_X = V_Y = 0.1\text{ V}$. Find value of V_O , I_{DX} , I_{DY} , I_{DL} .
(c)	What should be the operating mode of M_X and M_Y if $V_X = 1.8\text{V}$ and $V_Y = 0\text{ V}$? (Use logical reasoning, no need for calculation)



Solution:

(a)



$$V_{TNL} = -0.6V$$

$$V_{TND} = 0.4V$$

$$k_D = k'_D * \left(\frac{W}{L}\right)_D$$

$$= 133 \mu A/V^2$$

$$k_L = k'_L * \left(\frac{W}{L}\right)_L$$

$$= 100 \mu A/V^2$$

$$\therefore I_L = I_{D1} + I_{D2} = 2 I_{D1}$$

(*) M_L in Saturation $\rightarrow V_{GD} = (\text{small value}) - 1.8$

$$\therefore V_{GD} < V_{TNL}$$

Assume

(*) M_{D1} in Triode $\rightarrow V_{GD} = 1.8 - (\text{small value})$

$$\therefore V_{GD} > V_{TND}$$

$$\therefore k_L [(V_{GS} - V_{TNL})^2] = 2 \times k_{ND} [2(V_{GS} - V_{TND})V_{DS} - V_{DS}^2]$$

$$\rightarrow 100[0+0.6]^2 = 2 \times 133 [2(1.8-0.4)V_o - V_o^2]$$

$$\hookrightarrow V_o = \begin{cases} 0.0492 \rightarrow \text{Ans.} \\ 2.7508 \rightarrow V_o \neq V_{DD} \end{cases}$$

$$\therefore V_o = \boxed{0.0492 \text{ V.}}$$

$$\begin{aligned} \therefore I_{DL} &= k_L [V_{GSL} - V_{TNL}]^2 \\ &= (100 \mu\text{A}) [0.6]^2 = \boxed{3.6 \times 10^{-5} \text{ A}} \end{aligned}$$

$$\therefore I_{DX} = I_{DY} = \frac{I_{DL}}{2} = \boxed{1.8 \times 10^{-5} \text{ A.}}$$

$$\begin{aligned} \therefore \text{Power} &= I_L (V_{DD} - 0) = (3.6 \times 10^{-5}) (1.8) \\ &= \boxed{0.0648 \text{ m W.}} \end{aligned}$$

$$\text{Verify } M_L \rightarrow V_{GL} = V_{SL} = V_o = 0.0492 \text{ V}$$

$$\therefore V_{GD} = 0.0492 - 1.8 < V_{TNL} \quad (-0.6 \text{ V})$$

$$\textcircled{b} V_x = V_y = 0.1 \text{ V}$$

$$\therefore V_{GSX} = 0.1 - 0 = 0.1 \text{ V} < V_{TND} (0.4 \text{ V})$$

$$V_{GSY} = 0.1 - 0 = 0.1 \text{ V} < V_{TND}$$

$\therefore M_x, M_y$ OFF.

However, For $M_L \rightarrow V_{GSL} = 0 > V_{TNL} (-0.6 \text{ V})$

$\therefore M_L$ is ON. and

$$I_{DX} = I_{DY} = I_{DL} = \boxed{0 \text{ A.}}$$

$$\therefore V_o = V_{DD} = \boxed{1.8 \text{ V.}}$$

③ $V_x = 1.8$ and $V_y = 0V$

$$\therefore V_{GSY} = 0 - 0 = 0V < V_{TND} \quad (0.4V)$$

$\therefore M_y$ off.

For $M_x \rightarrow V_{GSX} = 1.8 > V_{TND}$

$\therefore M_x$ ON.

For NOR gate, if one input is high, output should be low.

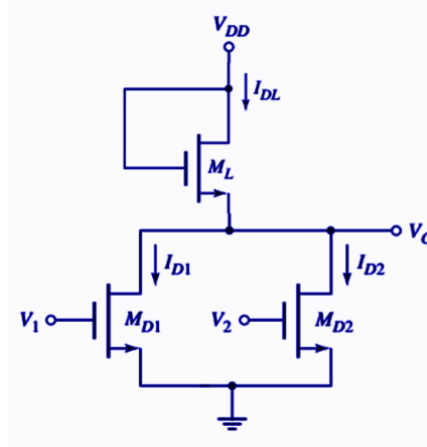
$$\therefore V_{GDx} = 1.8 - (\text{small value})$$

$\therefore V_{GDx}$ should be greater than V_{TND} .

$\therefore M_x$ should be in Triode region.

Question 3

The enhancement-load NMOS NOR gate in the figure is biased at $V_{DD} = 5.5 \text{ V}$. The transistor parameters are $K_n = 0.2 \text{ mA/V}^2$, $V_{TN1} = 0.4 \text{ V}$, $V_{TN2} = 1 \text{ V}$, $V_{TNL} = 0.9 \text{ V}$.



(a)	Find the operating mode of the load NMOS transistor.
(b)	Find the value of v_o in V and I_{DL} , I_{D1} , I_{D2} in mA for $v_1 = 5.5 \text{ V}$, $v_2 = 5.5 \text{ V}$
(c)	Find the operating mode of the transistor M_{D1} .

Solution:

a) $V_{DS} \geq V_{GS} - V_{TH} \rightarrow V_{GS} = V_{DS}$
 [saturation] As, $V_{DS} > V_{DS} - V_{TNL}$; M_L always in saturation.

b) $V_1 = V_2 = 5.5V$

$$I_{DL} = I_{D1} + I_{D2}$$

$$\Rightarrow K_n (V_{GS} - V_{TNL})^2 = K_n \left[2 \cdot (V_{GS_{D1}} - V_{TNL}) V_{DS_{D1}} - V_{DS_{D1}}^2 \right] + K_n \left[2 \cdot (V_{GS_{D2}} - V_{TNL}) V_{DS_{D2}} - V_{DS_{D2}}^2 \right]$$

$$\Rightarrow (V_{DD} - V_0 - V_{TNL})^2 = [2(5.5 - 0.4)V_0 - V_0^2] + [2(5.5 - 1)V_0 - V_0^2]$$

$$\Rightarrow (5.5 - V_0 - 0.9)^2 = 11V_0 - 0.8V_0 - V_0^2 + 11V_0 - 2V_0 - V_0^2$$

$$\Rightarrow (4.6 - V_0)^2 = 19.2V_0 - 2V_0^2$$

$$\Rightarrow 21.16 - 9.2V_0 + V_0^2 - 19.2V_0 + 2V_0^2 = 0$$

$$\Rightarrow 3V_0^2 - 28.4V_0 + 21.16 = 0$$

$$\Rightarrow V_0 = 8.65 \text{ V} > V_{DD}$$

$$I_{D1} = 2.86 \text{ mA}$$

$$V_0 = 0.8153 \text{ V}$$

$$I_{DD1} = 1.53 \text{ mA}$$

$$I_{DD2} = 1.3345 \text{ mA}$$

c) $V_{DS_{D1}} = 0.8153 \text{ V} < 4.5 \text{ V}$
 [triode]

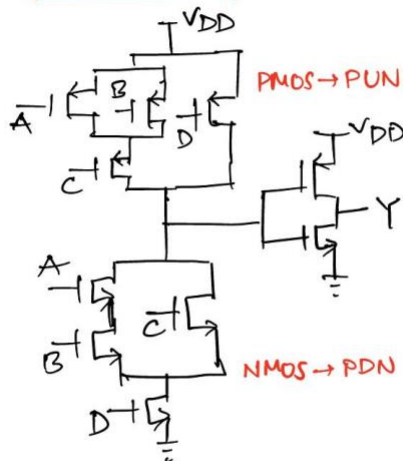
CMOS Logic

Question 4

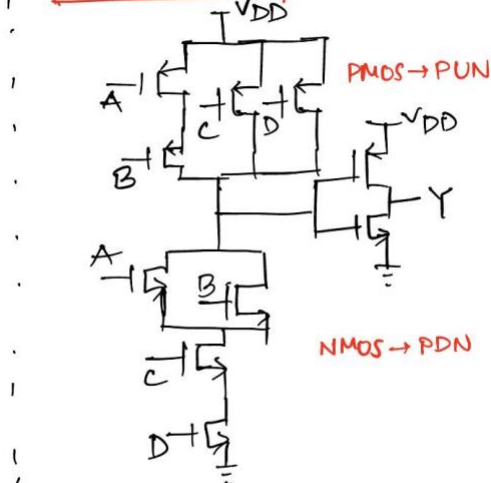
(a)	Design a static CMOS logic circuit that implements the logic function $Y = (AB+C)D$
(b)	Design a static CMOS logic circuit that implements the logic function $Y = (A+B)CD$
(c)	Design a static CMOS logic circuit that implements the logic function $Y = F(AB+C(D+E))$

Solution:

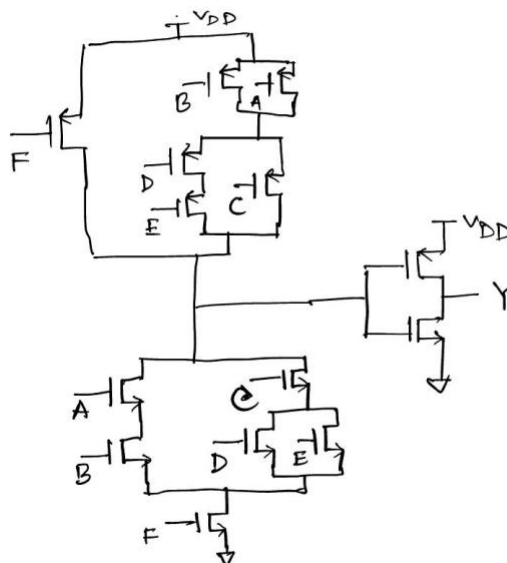
a) $Y = (AB+C)D$



b) $Y = (A+B)CD$

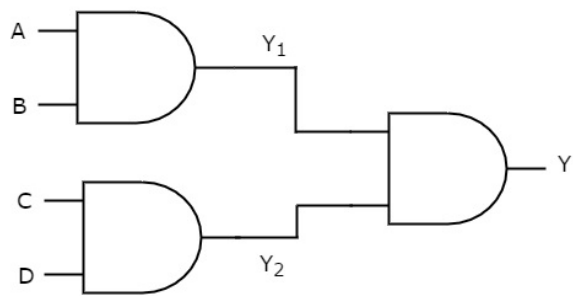


c) $Y = F(AB+C(D+E))$



Question 5

Design a CMOS logic circuit to implement the given compound gate in Figure below. First derive the logical expression of output Y and then design the CMOS network.



Solution:

$$Y = \text{AND}(Y_1, Y_2)$$

$$= \text{AND}(A \cdot B, C \cdot D)$$

$$= A \cdot B \cdot C \cdot D$$

